

Introduction

High energy costs and climate change have made energy consumption monitoring increasingly important. An Energy Consumption Monitoring System is an embedded system that measures and analyses electricity usage in real time, helping users reduce energy waste and lower electricity costs.

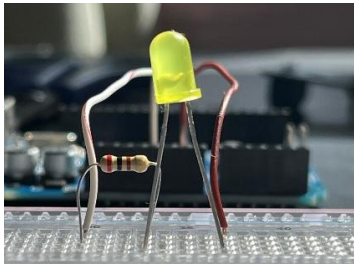
In Mauritius, electricity generation reached approximately 3,418 GWh in 2024, a 4.7% increase from 2023, while around 82% of electricity is still generated from non-renewable sources.

Figures highlight the importance of improving energy efficiency and promoting responsible electricity consumption. (Statistics Mauritius, 2025).

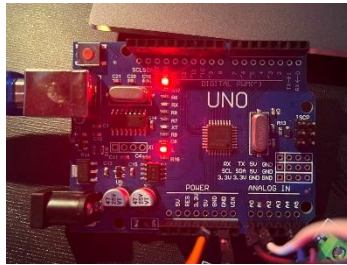
Inspired by commercial systems such as IAMMETER, a simplified Energy Consumption Monitoring System was developed.

Components Used

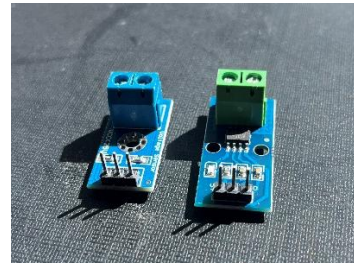
To keep costs low, the system was built using a simple setup: an Arduino Uno microcontroller, a basic current and voltage sensor, an LCD screen, and a simple circuit consisting of a resistor and an LED.



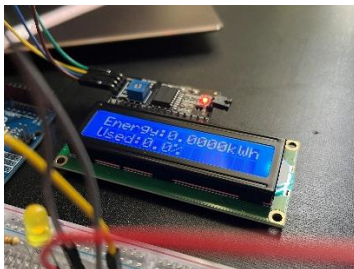
1. Simple circuit (Resistor and LED)



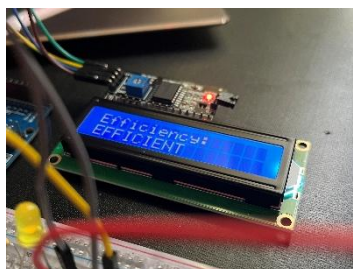
2. Microcontroller (Arduino UNO)



3. Current and Voltage Sensor



4. LCD screen (Showing Energy consumption per hour)



5. LCD screen (Showing whether system is efficient)

Microcontroller: The Arduino Uno acts as the brain of the system, handling real-time data sampling and running the control logic.

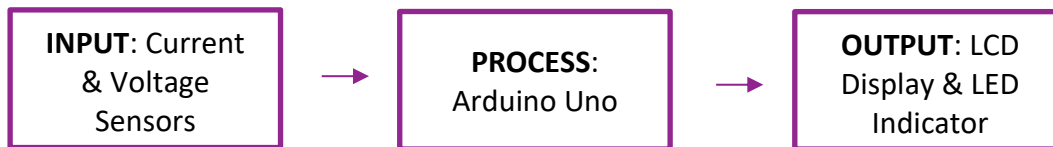
Current Sensor: It safely and automatically measures the electrical current drawn by the load and transmits this data to the Arduino board.

Voltage Sensor Module: It is the automatic sampling of voltage for processing by the Arduino Uno.

LCD display: It gives the user an instant look at real-time energy stats.

LED and Resistor: It acts as a visual indicator to show when the system is actively monitoring.

Working Principle



Input: The voltage and current sensors continuously monitor the system's analogue waveforms.

Process: The Arduino Uno's ADC samples the inputs to calculate instantaneous power ($P=IV$) and cumulative consumption (kWh). It then determines a usage percentage against a pre-programmed baseline of 8–15 kWh per day, based on the IEA global average threshold.

Output: The system outputs data locally to the LCD display, updating the message based on how hourly usage compares to the pre-programmed benchmark.

- Below 75%: Displays "Efficient Power".
- 75% to 100%: Displays "Warning: Near Limit".
- Above 100%: Displays "Limit Exceeded".

Benefits to users and society

Users:

- Financial Savings: Spotting standby waste prevents unnecessary electricity costs.
- User Empowerment: Real-time data shows exactly how daily habits affect bills.
- Home Protection: Catching sudden power surges early prevents electrical hazards and keeps people safe.

Society:

- Lowering national costs; less energy waste means the country spends less on expensive imported fuel, boosting the local economy.
- Stabilizing the grid; fewer peak-hour spikes mean reliable electricity for schools, hospitals, and homes with fewer blackouts.
- Protecting the environment; cutting overall power demand reduces carbon emissions, preserving local ecosystems and cleaner air for everyone.

Safety Considerations

The monitoring platform operates safely on a low voltage 5V DC circuit powered by the Arduino. Instead of direct electrical contact, the sensor uses the Hall-effect to measure current magnetically, creating a protective barrier that blocks high voltage from reaching the logic components. For prototyping, this setup is wired to a basic LED, resistor, and an LCD screen on a breadboard. For field deployment, a non-conductive enclosure is required to protect the exposed wiring.

Inclusion and accessibility aspects

The system replaces complex, technical utility meters with an intuitive, easy-to-read LCD screen, making energy data clear to a diverse demographic, including elderly users and children. It maintains strict economic accessibility by utilizing standard, cost-effective components so that lower-income households can afford the technology. Future iterations can integrate a dedicated mobile application to introduce assistive smartphone tools for individuals with varying levels of physical or technological literacy.

Environmental sustainability considerations

By providing real-time visibility into power consumption, the system drives conscious behavioral shifts toward active conservation, such as turning off idle devices. This reduction in household electricity usage directly lowers carbon emissions at the power generation level. Furthermore, the monitoring device features a low-power design; the microcontroller utilizes firmware-driven deep-sleep modes between sampling intervals, keeping its own operational footprint minimal.

Possible Improvements

Future upgrades could consist of the addition of an ESP8266 Wi-Fi module to send hourly or daily energy reports to a mobile app. This app can feature a game-like setup with rewards and challenges to encourage green habits. Additionally, adding a relay module and lightweight AI (TinyML) could allow the system to analyze appliance data and automatically cut power to low-priority electronics if the household gets too close to the 15-kWh daily limit.

Conclusion

The Energy Consumption Monitoring System demonstrates how engineering can be aligned with Public Interest Design principles. By providing safe, affordable, and easy-to-read energy data, it addresses both economic and environmental needs. Upgrading the system with lightweight AI (TinyML) shifts it from a passive display to an active tool, empowering users to cut costs and manage community resources more sustainably.